

Network topology refers to how devices in a computer network are connected. Network Topology is the way that defines the structure, and how these components are connected. Network Topology is divided into two types:

- **Physical Topology:** Physical topology indicates the arrangement of different elements of a network. It reflects the physical layout of devices and cables to form a connected network.
- **Logical Topology:** Logical topology defines the path that data packets take as they travel through the network independent of its physical layout.

Basic Terminology

- **Node:** A node refers to any device or point in the network, such as computers, [servers](#), [routers](#), or [switches](#). These nodes are crucial for transmitting and receiving data within the network.
- **Link:** A link signifies a physical or logical connection between two nodes, allowing data to flow between them. Links can be wired ([Ethernet cables](#)) or wireless ([Wi-Fi](#)).
- **Topology:** Topology is the arrangement or layout of nodes and links in a network. It determines how devices are interconnected and how data travels within the network.

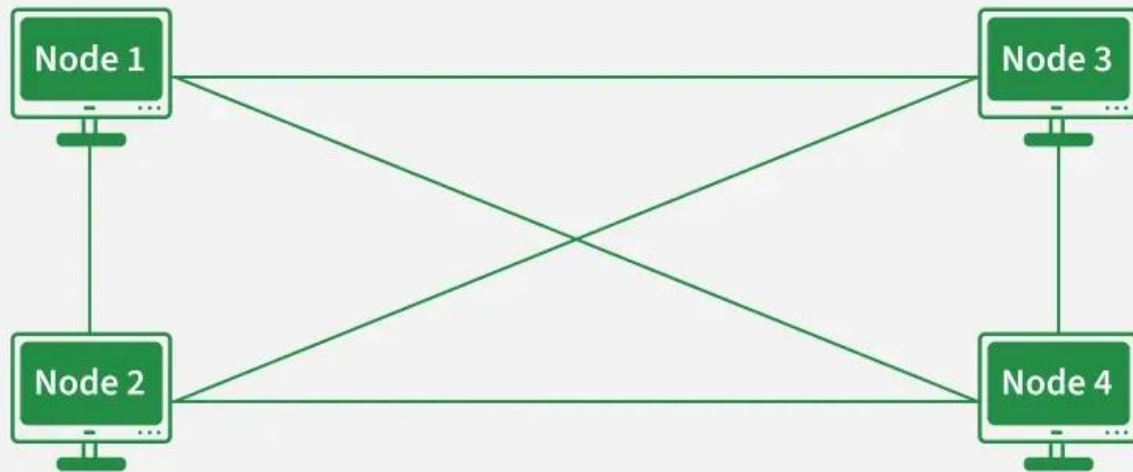
Types of Network Topology

The arrangement of a network that comprises nodes and connecting lines via sender and receiver is referred to as Network Topology. Below are various network topologies that include:

1. Mesh Topology

In [Mesh Topology](#), every node has a dedicated point-to-point link in every other node. Such a network is called complete because, for any two devices, there is a special link and non-redundant links cannot be added to the main network.

Mesh Topology



- Suppose, the N number of devices are connected with each other in a mesh topology, the total number of ports that are required by each device is N-1. There are 5 devices connected to each other, hence the total number of ports required by each device is 4. The total number of ports required = $N * (N-1)$.
- Suppose, N number of devices are connected with each other in a mesh topology, then the total number of dedicated links required to connect them is NC_2 i.e. $N(N-1)/2$. In Figure 1, there are 5 devices connected to each other, hence the total number of links required is $5*4/2 = 10$.

There are two types of Mesh topologies:

- **Full Mesh Topology** : All the nodes within the network are connected with every other. If there are n number of nodes during a network, each node will have an n-1 number of connections.
- **Partial Mesh Topology** : The partial mesh is more practical as compared to the full mesh. In a partially connected mesh, all the nodes aren't necessary to be connected with one another during a network.

Advantages of Mesh Topology

- Easy fault identification and isolation.
- Failure during a single device won't break the network.

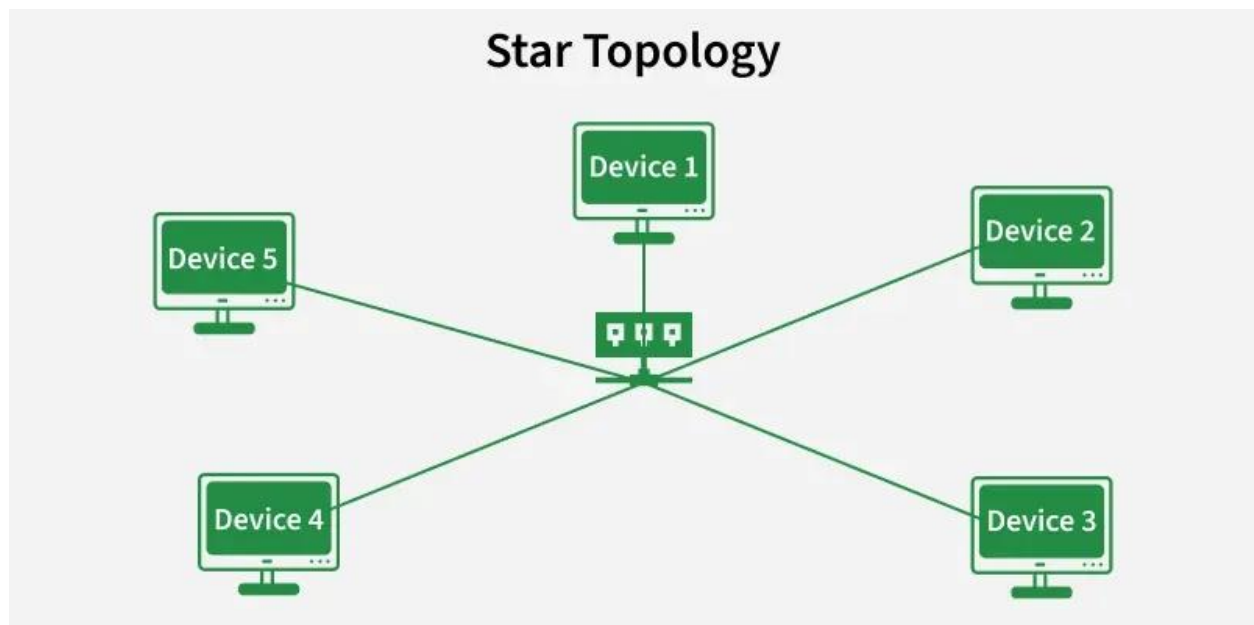
- There is no traffic problem as there is a dedicated point to point links for every computer.
- It provides high privacy and security.
- [Data transmission](#) is more consistent because failure doesn't disrupt its processes.
- Adding new devices won't disrupt data transmissions.
- This topology has robust features to beat any situation.
- A mesh doesn't have a centralized authority.

Disadvantages of Mesh Topology

- Each node must have an interface for every other node.
- There is only a limited number of I/O ports in a computer.
- The cost to implement mesh topology is high
- There is a high risk of redundant connections.
- Maintenance needs are challenging with a mesh topology.

2. Star Topology

In a [Star Topology](#), all the nodes (PCs, printers and [peripherals](#)) are connected to the central server. It has a central connection point, like a hub or switch. In star topology .each device is connected with central hub.



Advantages of Star Topology

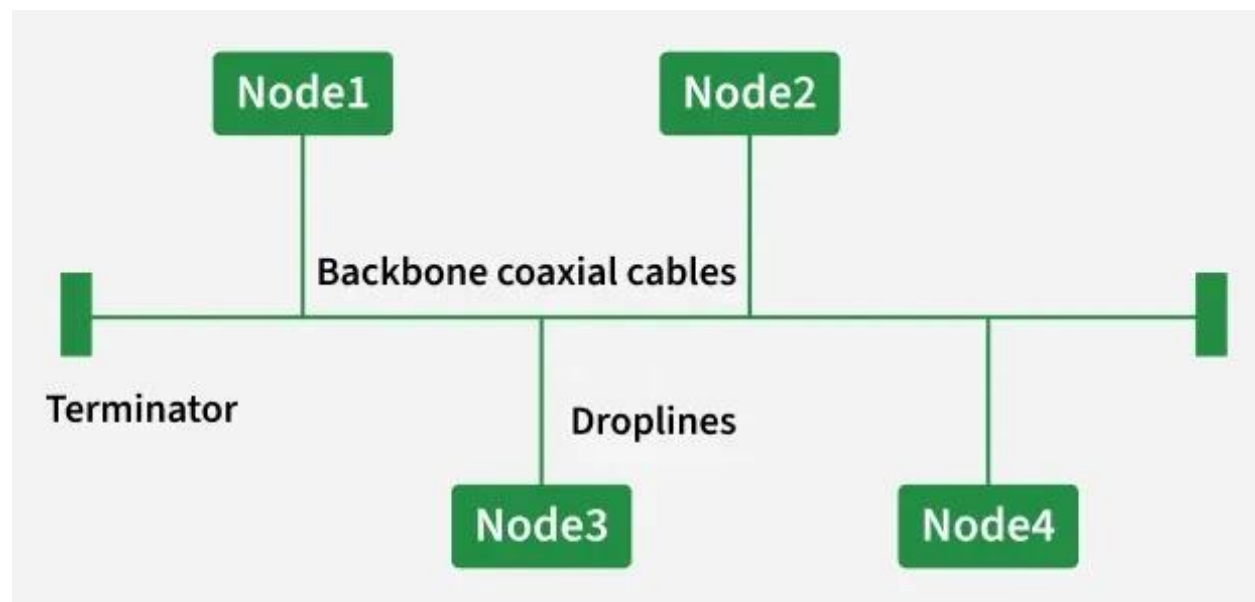
- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N. So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N.
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Disadvantages of Star Topology

- Star networks can require more cable length than a linear topology.
- More expensive cabling.
- Performance is based on the single concentrator i.e. hub.

3. Bus Topology

In [bus topology](#), all stations are attached to the same cable. In the bus network, messages are sent to both directions from a single point. In the bus topology, signals are broadcast to all stations. Each computer checks the address on the signal (data frame) as it passes along the bus. If the signal's address matches that of the computer, the computer processes the signal. If the address doesn't match, the computer takes no action and travels down the bus.



Advantages of Bus Topology

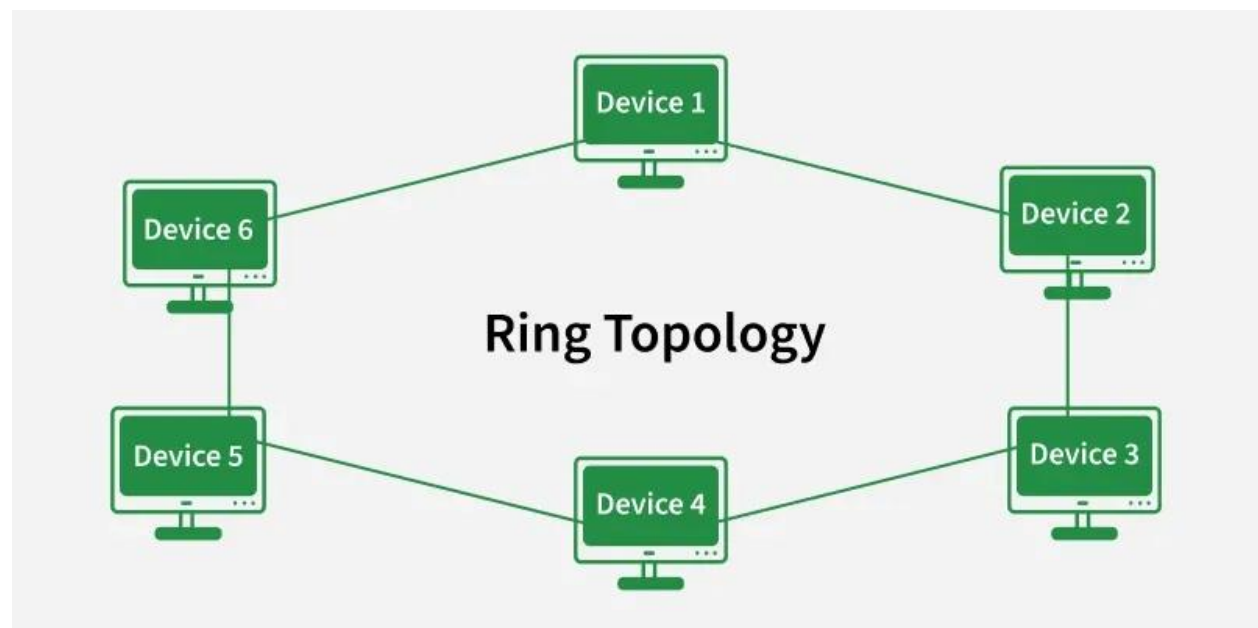
- If N devices are connected to each other in a bus topology, then the number of cables required to connect them is 1, known as backbone cable, and N drop lines are required.
- Coaxial or [twisted pair cables](#) are mainly used in bus-based networks that support up to 10 Mbps.
- The cost of the cable is less compared to other topologies, but it is used to build small networks.

Disadvantages of Bus Topology

- A bus topology is quite simpler, but still, it requires a lot of cabling.
- If the common cable fails, then the whole system will crash down.
- If the network traffic is heavy, it increases collisions in the network. To avoid this, various protocols are used in the MAC layer known as [Pure Aloha](#), [Slotted Aloha](#), [CSMA/CD](#), etc.
- Adding new devices to the network would slow down networks.
- Security is very low.

4. Ring Topology

All the nodes in a [Ring Topology](#) are connected in a closed circle of cable. Messages that are transmitted travel around the ring until they are addressed to, the signal being refreshed by each node. In a ring network, every device has exactly two neighbors for communication purposes.



The most common access method of ring topology is [token passing](#).

- **Token passing:** It is a network access method in which a token is passed from one node to another node.
- **Token:** It is a frame that circulates around the network.

Advantages of Ring Topology

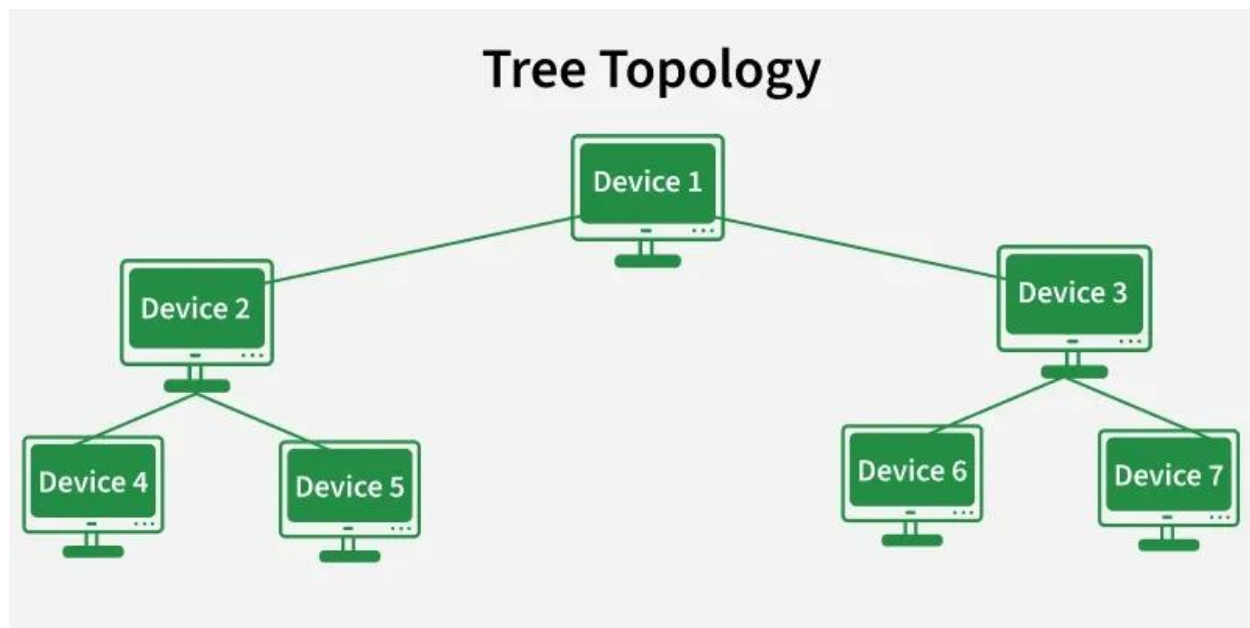
- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.
- It is less costly than a star topology.

Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- The addition of stations in between or the removal of stations can disturb the whole topology.
- Less secure.

5. Tree Topology

In [tree topology](#) nodes are connected in a hierarchical structure to form a tree. There is a root node in tree topology and the remaining nodes are considered as child nodes, basically it is a combination of star and bus topology. The central bus works as a communication pathway, and each star-configured network represents a level in the tree. In tree topology, a hierarchy is formed by the branching cable having no loops that connect the root with all other nodes for communication.



Advantages of Tree Topology

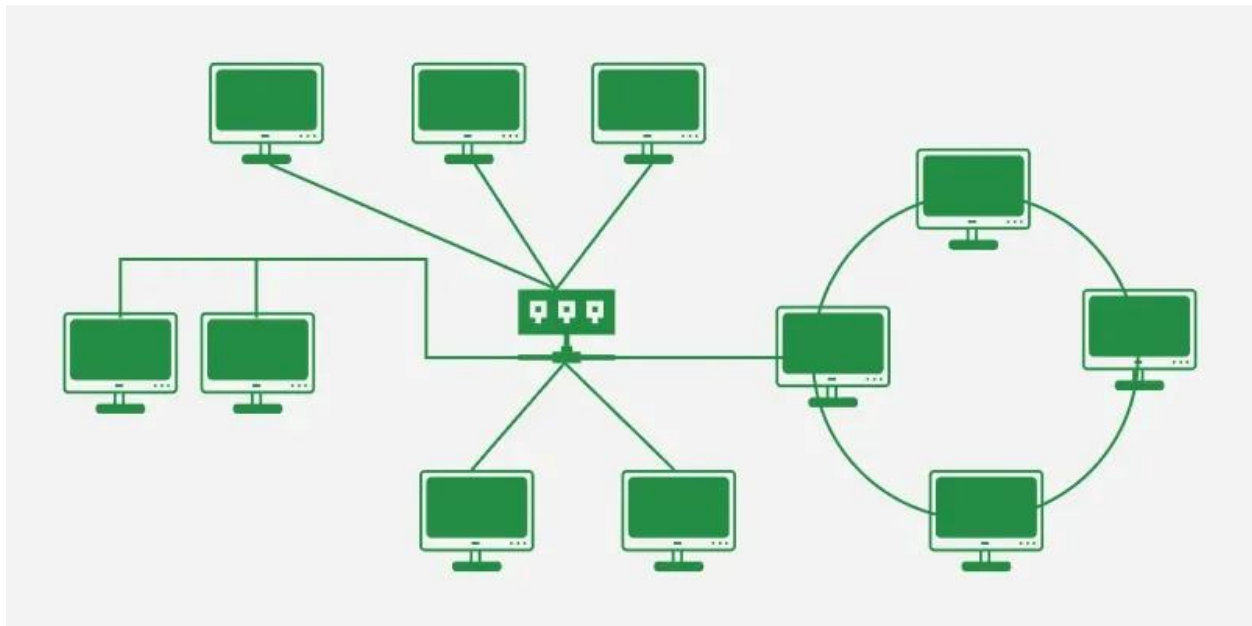
- Security is high in Tree Topology
- Tree Topology is more reliable
- Tree topology is more scalable
- It allows more devices to be attached to a single central hub thus it decreases the distance that is traveled by the signal to come to the devices.
- We can add new devices to the existing network.
- [Error detection](#) and [error correction](#) are very easy in a tree topology.

Disadvantages of Tree Topology

- If the central hub gets fails the entire system fails.
- The cost is high because of the cabling.
- If new devices are added, it becomes difficult to reconfigure.

6. Hybrid Topology

[Hybrid topology](#) is the combination of two or more types of topology, they arise from the integration of multiple network topologies that is why called Hybrid Network Topology.



Advantages of Hybrid Topology

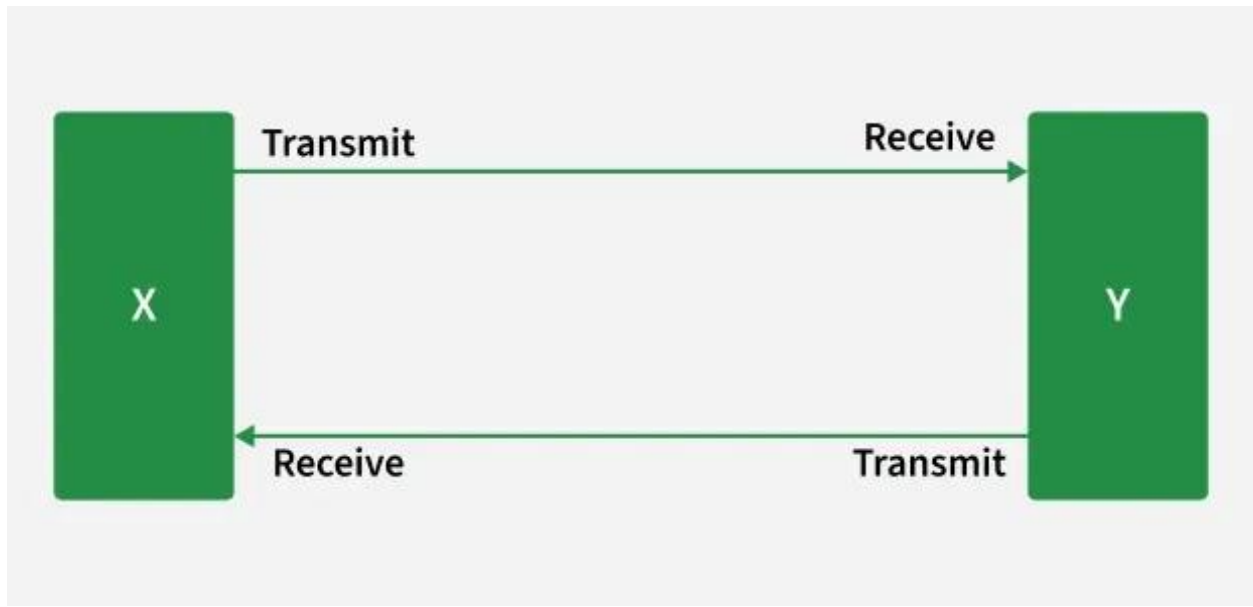
- Hybrid Topology provides more flexibility than others topologies.
- Hybrid topology is more robust than the other.
- Hybrid Topology provides optimized performance

Disadvantage of Hybrid Topology

- It is challenging to design the architecture of the Hybrid Network.
- Hubs used in this topology are very expensive.
- The infrastructure cost is very high as a hybrid network requires a lot of cabling and network devices.

7. Point to Point Topology

Point-to-Point Topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.



Advantages of Point to Point Topology

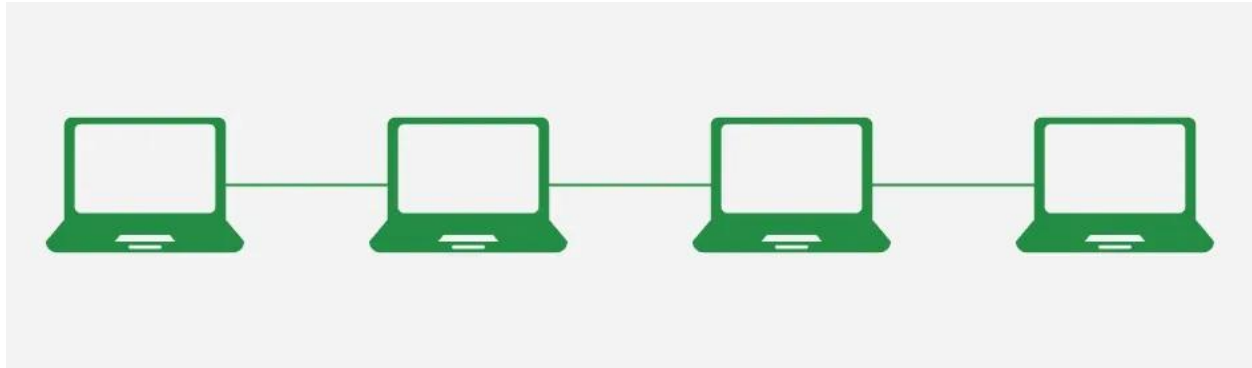
- P2P networks are highly efficient as they allow for direct communication between two devices without any intermediate devices or network components.
- P2P networks are relatively more secure than other topologies as they do not rely on intermediate devices that can be compromised or attacked.
- P2P networks are easy to configure and require minimal management or administration.

Disadvantages of Point to Point Topology

- P2P networks are not scalable as adding new devices requires establishing a separate link between each new device and the existing network, which can be time-consuming and expensive.
- P2P networks can be difficult to maintain as each device has to be managed separately.
- P2P networks do not provide redundancy, which can be a problem if a link fails or a device goes offline. with a dedicated communication connection between two systems.

8. Daisy Chain Topology

The daisy chain topology connects nodes along a chain of connections. Data is transported from one node to the next until it reaches its intended destination. There are two types of daisy chain network topologies: linear daisy chains and ring daisy chains.



Advantages of Daisy Chain Topology

- Easy to create and maintain
- Less cable usage
- Cost-Effective

Disadvantages of Daisy Chain Topology

- It provide slow data transmission.
- Easy failure, particularly in linear daisy chain topologies
- Every nodes needs two transmitters and two receivers for successful data transmission.